

Chapter 10. System Sequence Diagram

“In theory, there is no difference between theory and practice. But, in practice, there is.”

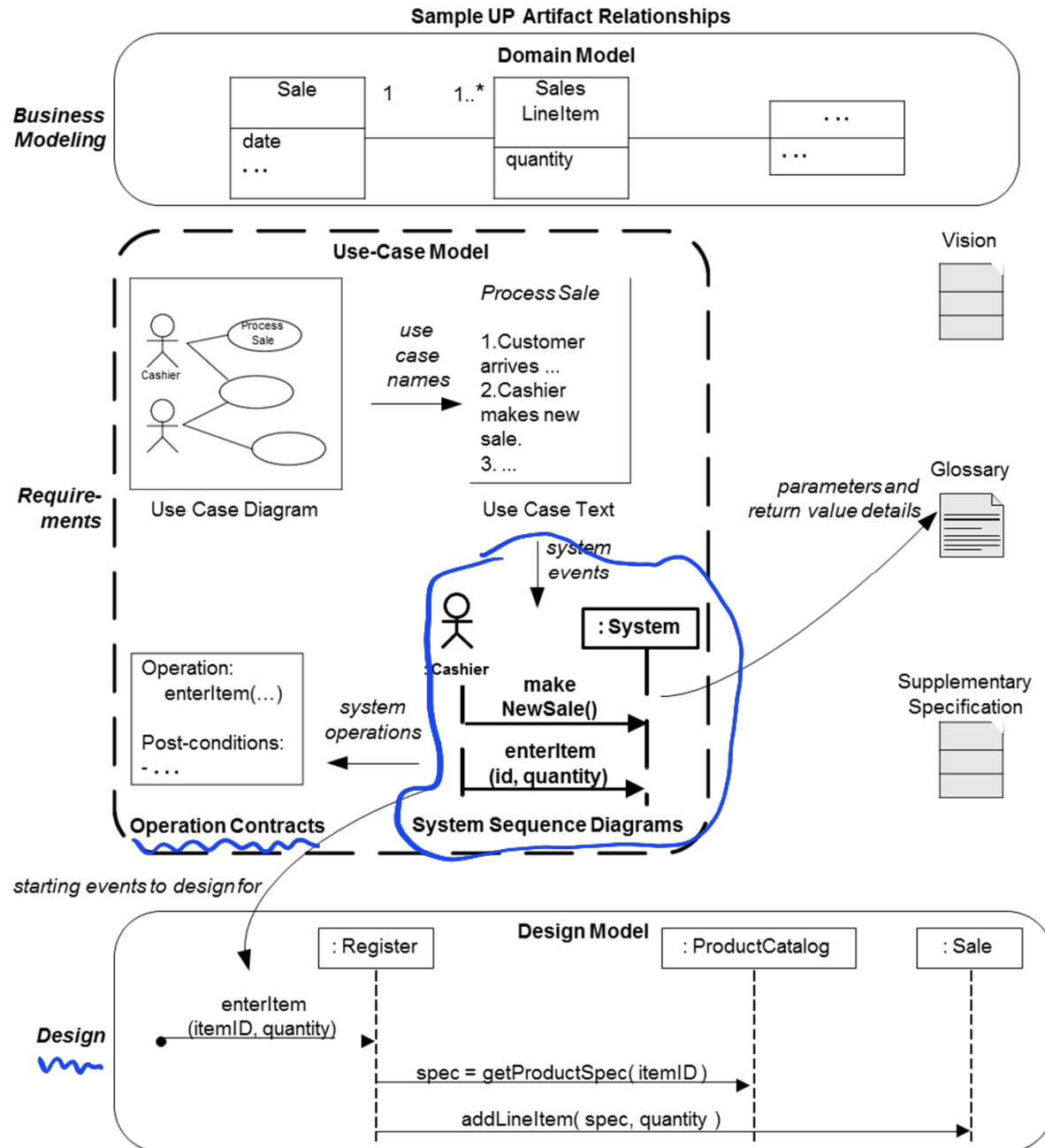
Objectives

- Identify system events.
- Create system sequence diagrams for use case scenarios.

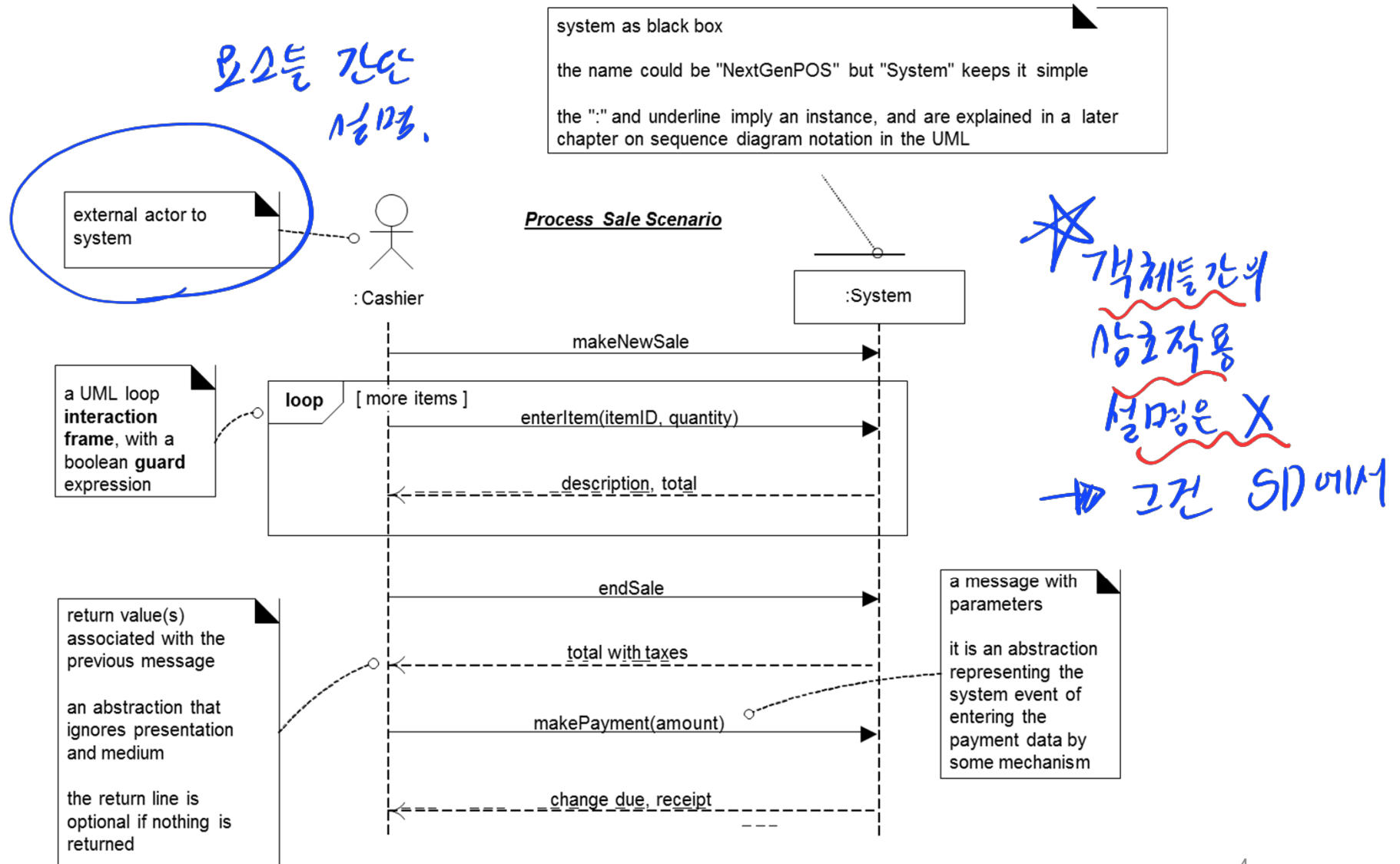
System Sequence Diagram

- A **system sequence diagram (SSD)** → 시스템과 관련 input & output event를 보여줌.
 - illustrates **input and output events** related to the system
 - used for deriving **operation contracts**
 - used for **object design**
 - UML sequence diagram → object design & operation contracts 작성시에 활용,
- An SSD shows, for a particular course of events within a use case,
 - **the external actors** that interact directly with the system,
 - **the system** (as a black box), and
 - **the system events** that the actors generate.
 - Time proceeds downward, and the ordering of events should follow their order in the scenario.

↳ 외부 actor, 시스템, 이벤트, interaction frame 등을 보여줌.



Example: An SSD for NextGen POS



SSD를 그리는 이유.

Why Draw an SSD?

- Useful to know precisely **what are the external input events** – the **system events**, to which the system should react.

↳ 시스템 이벤트와 외부 입력 이벤트를 명확하게 보여줌.

- System events include:

- **External events from actors** (humans or computers)
- **Timer events**
- **Faults or exceptions** (usually caused by external events)

- SSDs depict system behavior

- in terms of **what the system does**, not ~~how~~ it does it (remember: **blackbox!**)
- System behavior is also captured by use cases and operation contracts.

Process: Iterative and Evolutionary SSDs

- Most **SSDs** are created during **elaboration**, when it is useful to
 - identify the details of the **system events** to clarify what major operations which the system must be designed to handle,
 - write **system operation contracts**, and
 - possibly to support **estimation**.